

1. Project Overview:

The objective of this project was to clean a customer dataset by identifying and resolving data quality issues such as missing values, duplicate records, invalid values, and inconsistencies. After cleaning, an exploratory analysis was performed to gain initial business insights.

2. Data Cleaning Summary:

Task	Result
Missing Values	Handled
Duplicate Records	Removed
Invalid Age	Corrected
Gender Issues	Corrected
Inconsistent Records	Reviewed

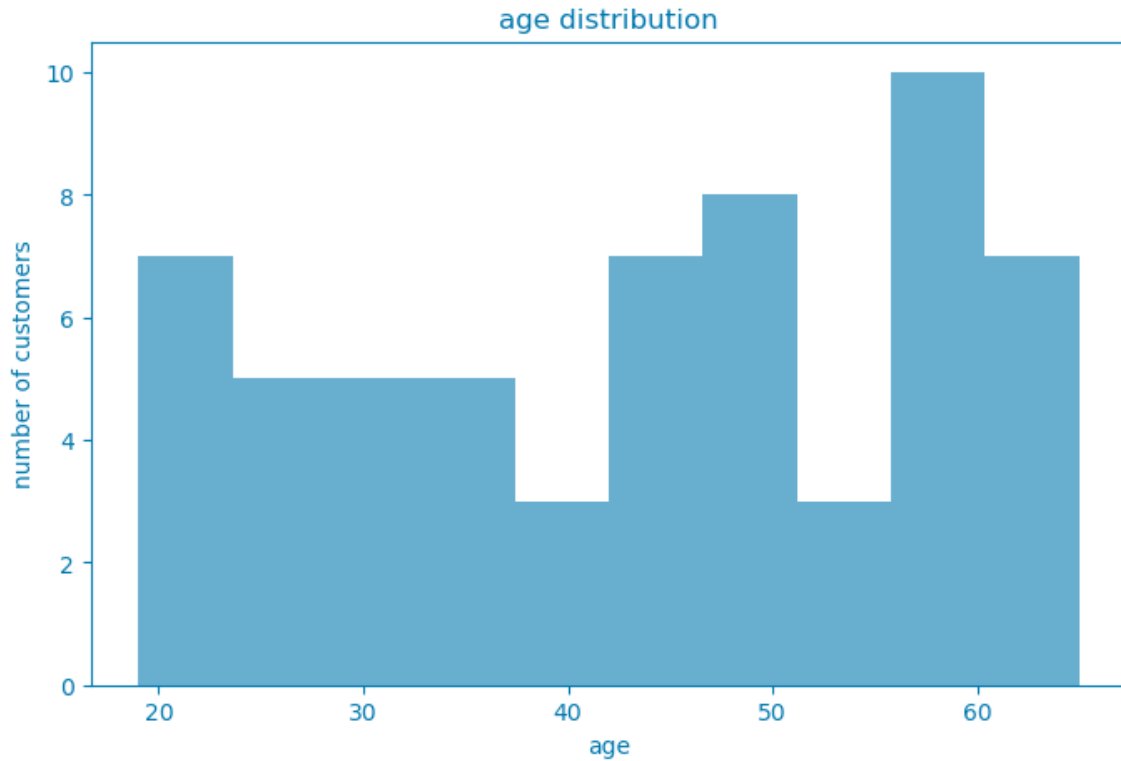
3. Age Distribution:

Analysis:

The age distribution indicates that customers are spread across a wide range of ages, from approximately 19 to 65 years. The highest concentration of customers falls within the 55-60 age group, while younger age groups are represented less frequently. After the data cleaning process, unrealistic age values were corrected, resulting in a more reliable age distribution.

Business Insight:

Since a large proportion of customers belong to the 55-60 age group, marketing strategies and promotional campaigns can be tailored to better meet the preferences of this customer segment. At the same time, attracting younger customers could help diversify the customer base and support long-term business growth.



4. Membership Tier Distribution:

Analysis:

The majority of customers belong to the Gold membership tier, while fewer customers are in the VIP and Silver tiers. This suggests that only a small portion of customers have reached higher loyalty levels.

Business Insight:

Encouraging customers to upgrade their membership through loyalty programs and personalized rewards may help increase customer retention and revenue.



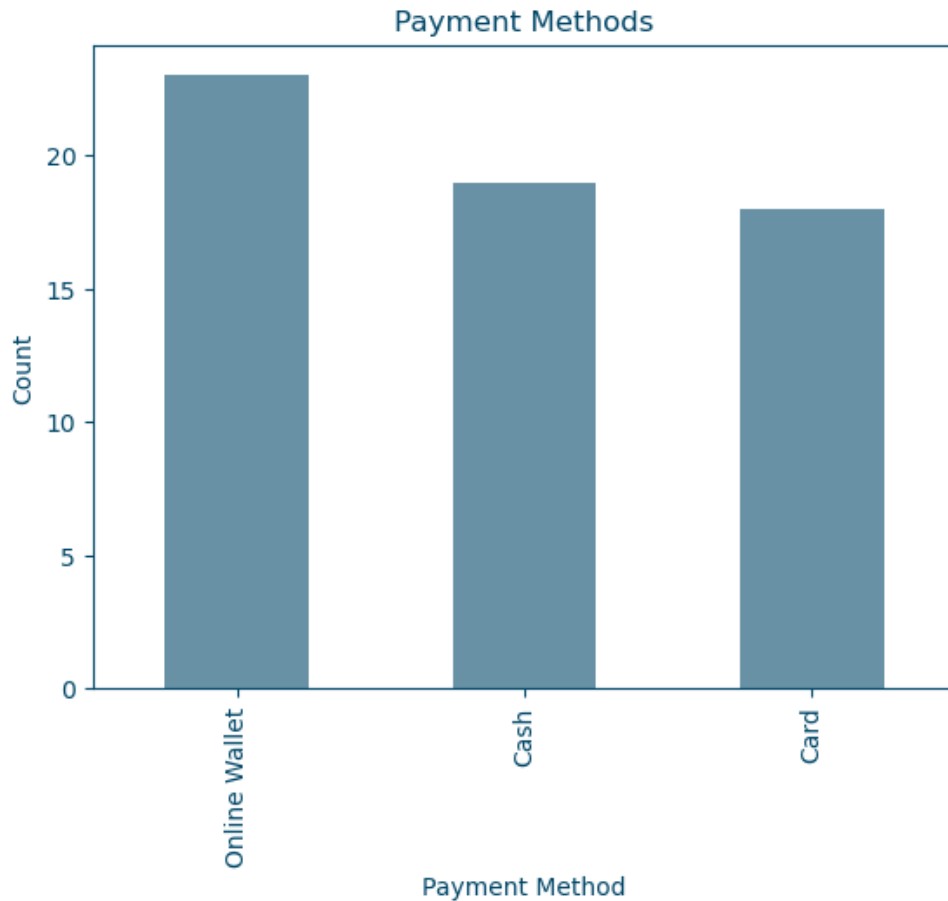
3. Payment Method Distribution:

Analysis:

The chart illustrates the distribution of customer payment methods. Some payment methods like online walet are used more frequently than others, indicating customer preferences during the checkout process.

Business Insight:

Understanding customer payment preferences can help businesses optimize payment options and improve the overall purchasing experience.



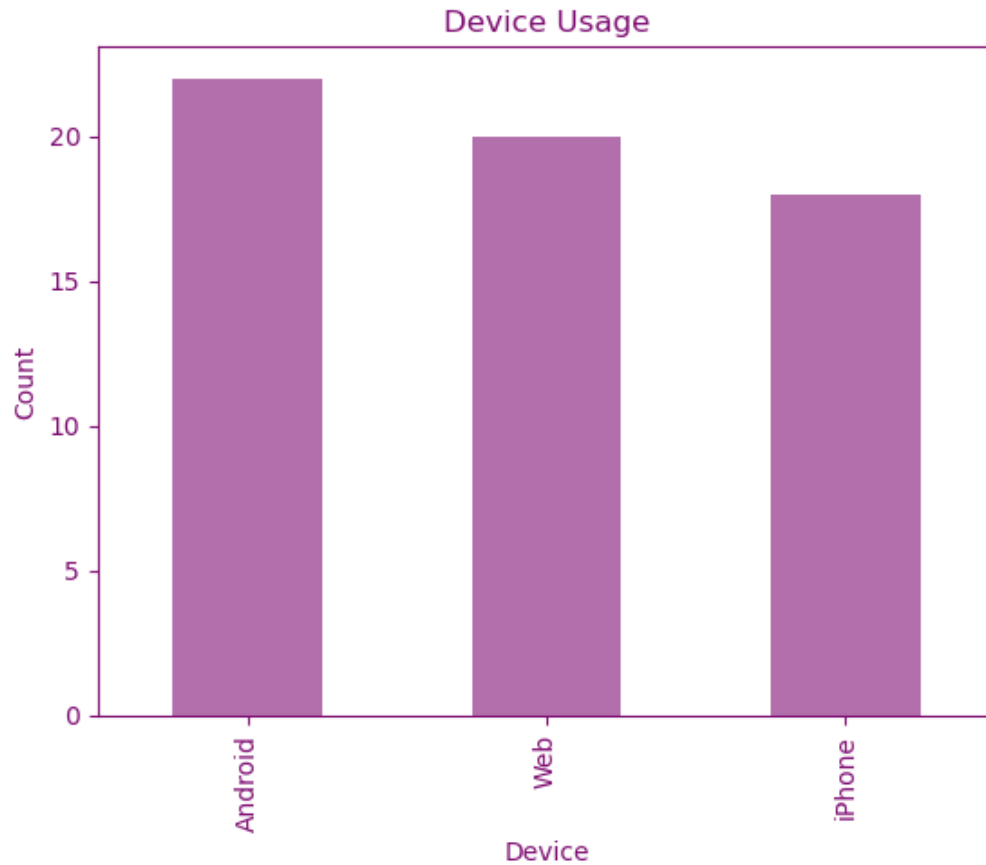
4. Device Usage

Analysis:

The device usage analysis shows that Android is the most commonly used platform, followed by Web and iPhone. This indicates that the majority of customers prefer accessing the platform through Android devices.

Business Insight:

Since Android represents the largest share of users, optimizing the Android application should be a priority. At the same time, maintaining a reliable web experience is important due to its significant user base, while continued support for iPhone users ensures accessibility across all major platforms.



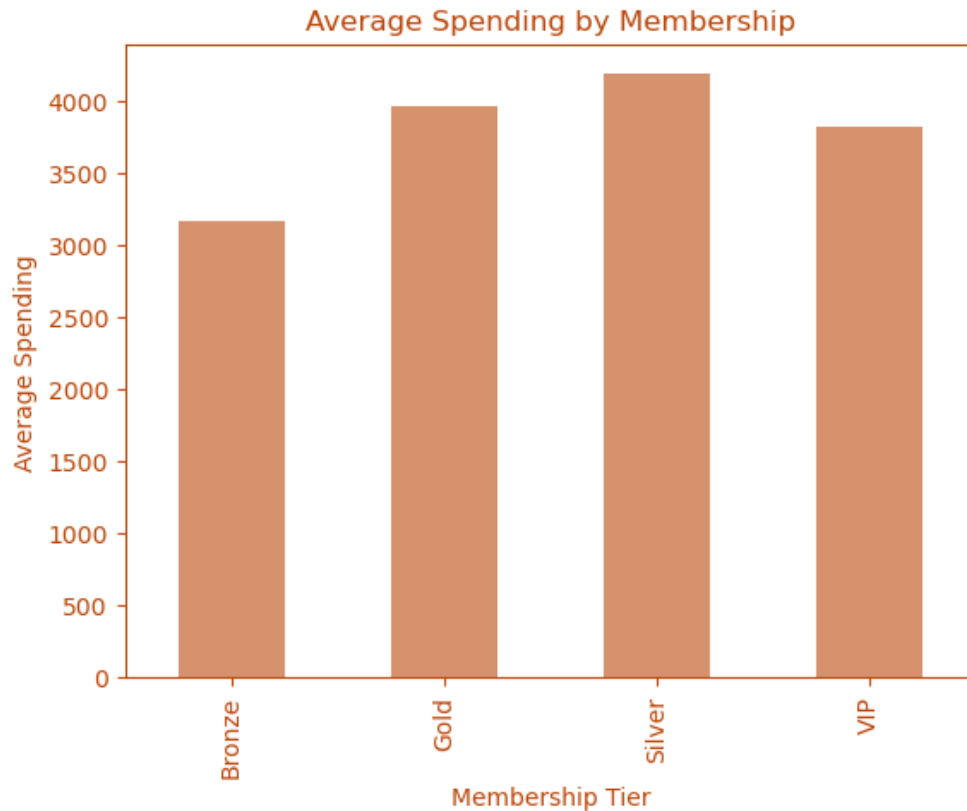
5. Average Spending by Membership Tier:

Analysis:

The analysis shows that Silver members have the highest average spending compared to the other membership tiers. This suggests that Silver customers are currently the most valuable customer segment in terms of average purchase value.

Business Insight:

Since Silver members demonstrate the highest spending behavior, the business could focus on retaining these customers while encouraging them to upgrade to Gold or VIP through targeted loyalty programs and personalized offers



Conclusion

This project successfully improved the quality of the customer dataset through a structured data cleaning process. Missing values were handled, duplicate records were removed, invalid values were corrected, and data inconsistencies were identified and reviewed. As a result, the dataset became more accurate and reliable for further analysis.

The exploratory data analysis revealed several useful business insights. Most customers use **Android** devices, **Silver** members have the highest average spending among all membership tiers, and the customer age distribution is concentrated around the middle to older age groups. These findings can help businesses better understand customer behavior and support data-driven decision-making.

Overall, the cleaned dataset is now ready for advanced analysis, dashboard development, or machine learning applications.